NCSC101: Practice Set - MCQ Questions

Comprehensive Practice Questions

Prof. Pranay Kumar Saha

Instructions:

- This practice set contains 40 multiple-choice questions.
- Each question has only one correct answer.
- Read each question carefully before answering.
- Solutions and explanations are provided after each question.

1. What is the output of the following Python code?

```
name = "Alice"
age = 30
info = f"{name} will be {age + 5} in five years."
print(info)
```

   A. Alice will be 30 + 5 in five years.
   B. Alice will be 35 in five years.
   C. name will be age + 5 in five years.
   D. SyntaxError

> **Solution: Explanation:** F-strings (formatted string literals) allow embedding expressions inside string literals using curly braces {}. The expression `age + 5` is evaluated (30 + 5 = 35) and inserted into the string. This is one of the most readable ways to format strings in Python.

2. Which of the following is a mutable data structure in Python?
   A. Tuple
   B. List
   C. String
   D. Integer

Solution: Explanation: Lists are mutable, meaning their contents can be modified after creation (e.g., using `append()`, `remove()`, or direct indexing). Tuples, strings, and integers are immutable—once created, their values cannot be changed.

3. What does the following code print?

```
path = r'C:\Users\new_folder'
print(path)
```

   A. C:\Users\new_folder
   B. C:Users ew_folder
   C. SyntaxError
   D. C:/Users/new_folder

Solution: Explanation: The `r` prefix creates a raw string where backslashes are treated as literal characters, not escape sequences. This is particularly useful for Windows file paths and regular expressions. Without the `r` prefix, `\n` would be interpreted as a newline character.

4. What is the result of the following operation?

```
print(17 // 3)
```

   A. 5.666666666666667
   B. 5
   C. 6
   D. 2

Solution: Explanation: The `//` operator performs floor division, which divides and rounds down to the nearest integer. `17 / 3 = 5.666...`, and floor division gives us 5. This is different from regular division (`/`) which returns a float.

5. Which statement about Python indentation is correct?
   A. Python uses curly braces for code blocks
   B. Python uses indentation to define code blocks (PEP 8 recommends 4 spaces)
   C. Indentation is optional in Python
   D. Python uses semicolons to mark code blocks

Solution: Explanation: Python uses indentation to define code blocks instead of curly braces or keywords. PEP 8 (Python's style guide) recommends using 4 spaces per indentation level. Incorrect indentation will cause an `IndentationError`.

6. What does `sys.argv[0]` contain in a Python script?
   - A. The first command-line argument
   - B. The name of the script itself
   - C. The Python interpreter path
   - D. An empty string

Solution: Explanation: `sys.argv` is a list containing command-line arguments. The first element (`sys.argv[0]`) is always the script name. Actual arguments passed to the script start from `sys.argv[1]` onwards. For example, running `python script.py arg1 arg2` gives `sys.argv = ['script.py', 'arg1', 'arg2']`.

7. Which `os` module function returns the current working directory?
   - A. os.chdir()
   - B. os.getcwd()
   - C. os.listdir()
   - D. os.path.dirname()

Solution: Explanation: `os.getcwd()` returns the Current Working Directory as a string. `os.chdir()` changes the directory, `os.listdir()` lists directory contents, and `os.path.dirname()` returns the directory name from a path.

8. What is the difference between `os.mkdir()` and `os.makedirs()`?
   - A. They are identical functions
   - B. `os.mkdir()` creates a single directory; `os.makedirs()` creates nested directories
   - C. `os.mkdir()` is deprecated
   - D. `os.makedirs()` only works on Windows

Solution: Explanation: `os.mkdir()` can only create a single directory and will fail if parent directories don't exist. `os.makedirs()` can create all intermediate parent directories in a path. For example, `os.makedirs('parent/child/grandchild')` creates all three directories if they don't exist.

9. Which function should you use to safely join file paths across different operating systems?
   - A. String concatenation with "/"
   - B. String concatenation with "\\"
   - C. os.path.join()
   - D. os.path.split()

> **Solution: Explanation:** `os.path.join()` automatically uses the correct path separator for the operating system (backslash on Windows, forward slash on Linux/macOS). Manual string concatenation with "/" or "\" breaks cross-platform compatibility.

10. In `argparse`, what does `action="store_true"` do?
    - A. Requires the user to provide a value
    - B. Makes the argument a boolean flag (True if present, False otherwise)
    - C. Stores the string "true"
    - D. Raises an error

> **Solution: Explanation:** `action="store_true"` creates a boolean flag. If the flag is present in the command line, the value is `True`; if absent, it's `False`. This is commonly used for options like `--verbose` or `--debug` that don't need values.

11. What does `nargs='+'` mean in `argparse`?
    - A. Zero or more arguments
    - B. One or more arguments (produces a list)
    - C. Exactly one argument
    - D. Optional argument

> **Solution: Explanation:** `nargs='+'` specifies that at least one argument is required, and multiple arguments can be provided. They are collected into a list. `nargs='*'` allows zero or more, `nargs='?'` allows zero or one.

12. Which `os.path` function checks if a path exists (file or directory)?
    - A. os.path.isfile()
    - B. os.path.isdir()
    - C. os.path.exists()
    - D. os.path.check()

Solution: Explanation: `os.path.exists()` returns `True` if the path exists, regardless of whether it's a file or directory. `os.path.isfile()` specifically checks for files, and `os.path.isdir()` checks for directories. There is no `os.path.check()` function.

13. What does the `git init` command do?
    - A. Clones a remote repository
    - B. Creates a new Git repository in the current directory
    - C. Commits changes
    - D. Pushes to remote

Solution: Explanation: `git init` initializes a new Git repository by creating a hidden `.git` directory in the current folder. This directory stores all version control information, including commits, branches, and configuration.

14. What is the purpose of the staging area in Git?
    - A. To permanently store commits
    - B. To prepare and review changes before committing
    - C. To backup files
    - D. To merge branches

Solution: Explanation: The staging area (or index) is an intermediate area where you can selectively prepare changes before creating a commit. It allows you to review changes with `git diff --staged`, create logical atomic commits, and separate completed work from work-in-progress.

15. Which command shows the difference between staged changes and the last commit?
    - A. git diff
    - B. git diff –staged
    - C. git status
    - D. git log

Solution: Explanation: `git diff --staged` (or `git diff --cached`) shows changes that have been staged for commit. `git diff` alone shows unstaged changes in the working directory. `git status` shows which files are modified/staged, and `git log` shows commit history.

16. What is the difference between a lightweight tag and an annotated tag in Git?
    A. There is no difference
    B. Lightweight tags are encrypted
    C. Annotated tags contain metadata (tagger, date, message); lightweight tags are just pointers
    D. Lightweight tags cannot be pushed to remote

> **Solution: Explanation:** Lightweight tags are simple pointers to a commit. Annotated tags (created with `-a`) are full objects stored in Git's database with tagger information, date, message, and can be signed. Annotated tags are recommended for releases because they contain more information.

17. In Git Flow branching strategy, what is the purpose of the `develop` branch?
    A. To hold production-ready code
    B. To serve as an integration branch for features
    C. To fix emergency bugs
    D. To prepare releases

> **Solution: Explanation:** In Git Flow, `develop` is the integration branch where features are merged for testing before release. `main` (or `master`) holds production code, `feature/*` branches are for new features, `hotfix/*` for emergency fixes, and `release/*` for release preparation.

18. What does `git clone` do?
    A. Creates a new branch
    B. Creates a local copy of a remote repository
    C. Merges branches
    D. Deletes a repository

> **Solution: Explanation:** `git clone <url>` creates a complete local copy of a remote repository, including all files, commits, branches, and history. It automatically sets up the remote repository as `origin` and checks out the default branch.

19. Which command is used to create and switch to a new branch in one step?
    A. git branch new-branch

B. git checkout -b new-branch

C. git merge new-branch

D. git switch new-branch

Solution: Explanation: `git checkout -b new-branch` creates a new branch and switches to it in one command. Alternatively, `git switch -c new-branch` does the same. `git branch new-branch` only creates the branch without switching, and `git switch new-branch` switches to an existing branch.

20. What does DVCS stand for, and what is its main advantage?

    A. Direct Version Control System; faster commits

    B. Distributed Version Control System; every developer has full repository history

    C. Dynamic Version Control System; automatic backups

    D. Decentralized Version Control System; no conflicts

Solution: Explanation: DVCS stands for Distributed Version Control System. Unlike centralized systems (like SVN), each developer has a complete copy of the repository with full history. This enables offline work, faster operations, better branching/merging, and natural redundancy.

21. What is the purpose of the `<!DOCTYPE html>` declaration?

    A. To define CSS styles

    B. To declare the document type and HTML version

    C. To import JavaScript

    D. To create a hyperlink

Solution: Explanation: The `<!DOCTYPE html>` declaration tells the browser that the document is an HTML5 document. It ensures the browser renders the page in standards mode rather than quirks mode. This should be the first line of every HTML document.

22. Which HTML tag is used for the largest heading?

    A. `<h6>`

    B. `<heading>`

    C. `<h1>`

    D. `<header>`

Solution: **Explanation:** `<h1>` represents the largest and most important heading. HTML provides six levels of headings from `<h1>` (largest) to `<h6>` (smallest). Each page should typically have only one `<h1>` tag for the main title.

23. What is the correct CSS syntax to change the font color of all paragraph elements to blue?
    - A. p {font-color: blue;}
    - B. `<p style="font: blue;">`
    - C. p {color: blue;}
    - D. p.color = blue

Solution: **Explanation:** The correct CSS property for text color is `color`, not `font-color`. The syntax is `selector {property: value;}`. This rule will apply to all `<p>` elements in the document.

24. Which CSS selector has the highest specificity?
    - A. Element selector (p)
    - B. Class selector (.classname)
    - C. ID selector (#idname)
    - D. Universal selector (*)

Solution: **Explanation:** Specificity determines which CSS rule applies when multiple rules target the same element. In order from lowest to highest: Universal selector (*), Element selector (p), Class selector (.class), ID selector (#id). Inline styles have even higher specificity, and !important overrides everything.

25. What does the CSS Box Model consist of (from innermost to outermost)?
    - A. Margin, Padding, Border, Content
    - B. Border, Content, Padding, Margin
    - C. Content, Padding, Border, Margin
    - D. Padding, Content, Margin, Border

Solution: **Explanation:** The CSS Box Model, from innermost to outermost, consists of: Content (the actual content area), Padding (space inside the border), Border (the border itself), and Margin (space outside the border). Understanding this is crucial for layout and spacing.

26. What is the purpose of the `alt` attribute in an `<img>` tag?
    - A. To specify the image width
    - B. To provide alternative text for accessibility and when image fails to load
    - C. To define image alignment
    - D. To set the image source

> **Solution: Explanation:** The `alt` attribute provides alternative text describing the image. It's displayed if the image fails to load, used by screen readers for accessibility, and helps with SEO. Always include meaningful alt text for images.

27. In the client-server architecture, what initiates the communication?
    - A. Server
    - B. Client
    - C. Database
    - D. Router

> **Solution: Explanation:** In client-server architecture, the client (e.g., web browser, mobile app) initiates communication by sending requests to the server. The server then processes the request and sends back a response. The client-server model is fundamental to web applications.

28. What does HTTPS provide that HTTP does not?
    - A. Faster data transfer
    - B. Encryption and secure communication using SSL/TLS
    - C. Better caching
    - D. Larger file uploads

> **Solution: Explanation:** HTTPS (HTTP Secure) encrypts data using SSL/TLS, protecting against eavesdropping and man-in-the-middle attacks. It ensures data integrity and authenticates the server. Modern browsers mark HTTP sites as "Not Secure," making HTTPS essential for all websites.

29. What protocol does the `ping` command use?
    - A. TCP
    - B. UDP

C. ICMP

D. HTTP

Solution: Explanation: `ping` uses ICMP (Internet Control Message Protocol), specifically ICMP Echo Request (Type 8) and Echo Reply (Type 0) messages. ICMP is a network layer protocol used for diagnostics and error reporting, not for data transfer.

30. What does a 0% packet loss in ping indicate?

A. The network is down

B. High latency

C. Good network connectivity

D. DNS failure

Solution: Explanation: 0% packet loss means all ping requests received replies, indicating stable connectivity between source and destination. Packet loss occurs due to congestion, hardware issues, or filtering. High packet loss (e.g., `>5%`) indicates network problems.

31. What is the default port for SSH?

A. 21

B. 22

C. 23

D. 80

Solution: Explanation: SSH (Secure Shell) uses port 22 by default. Port 21 is for FTP, port 23 for Telnet, and port 80 for HTTP. SSH provides encrypted remote access and file transfer, replacing insecure protocols like Telnet.

32. Which protocol is more secure for file transfer?

A. FTP

B. Telnet

C. SFTP

D. HTTP

Solution: Explanation: SFTP (SSH File Transfer Protocol) runs over SSH (port 22) and encrypts all data and commands. FTP transmits credentials and data in plain text. FTPS (FTP over SSL/TLS) is also secure, but SFTP is more commonly used and simpler to configure with firewalls.

33. What is the purpose of DNS?
    A. To encrypt web traffic
    B. To translate domain names to IP addresses
    C. To route packets
    D. To authenticate users

Solution: Explanation: DNS (Domain Name System) acts as the "phonebook" of the internet, translating human-readable domain names (like example.com) into machine-readable IP addresses (like 192.0.2.1). Without DNS, users would need to memorize IP addresses.

34. What is the main security advantage of SSH over Telnet?
    A. SSH is faster
    B. SSH encrypts all communication; Telnet sends data in plain text
    C. SSH uses less bandwidth
    D. SSH works on more operating systems

Solution: Explanation: SSH encrypts all communication including authentication credentials, protecting against eavesdropping and man-in-the-middle attacks. Telnet transmits everything in plain text, making credentials easily intercepted. SSH has completely replaced Telnet for remote access.

35. In FTP, which ports are used by default?
    A. 22 and 23
    B. 20 (data) and 21 (control)
    C. 80 and 443
    D. 25 and 110

Solution: Explanation: FTP uses two ports: port 21 for control commands (login, file operations) and port 20 for data transfer. This two-port design can cause firewall issues. SFTP uses only port 22 (SSH), making it simpler to configure.

36. What type of DNS record maps a domain name to an IPv4 address?
    A. A record
    B. AAAA record
    C. CNAME record
    D. MX record

> **Solution: Explanation:** An A record maps a domain name to an IPv4 address. AAAA records map to IPv6 addresses, CNAME records create aliases, MX records specify mail servers, and TXT records store text information (often for verification or email security).

37. What is a reverse proxy used for?
    A. Hiding client IP addresses
    B. Sitting in front of web servers to handle incoming requests (load balancing, caching)
    C. Encrypting DNS queries
    D. Blocking malware

> **Solution: Explanation:** A reverse proxy sits in front of web servers and handles incoming client requests. It can perform load balancing (distributing traffic across multiple servers), caching, SSL termination, and security filtering. Examples include Nginx and Apache as reverse proxies.

38. What does TTL stand for in networking, and what is its purpose?
    A. Total Traffic Load; measures bandwidth
    B. Time To Live; prevents infinite routing loops
    C. Transfer Time Limit; sets timeout
    D. Total Transmission Length; specifies packet size

> **Solution: Explanation:** TTL (Time To Live) is a field in IP packet headers that gets decremented by 1 at each router hop. When TTL reaches 0, the packet is discarded. This prevents packets from circulating indefinitely in routing loops. Traceroute uses TTL to discover network paths.

39. Which command would you use to test if port 443 is open on a server using telnet?
    A. telnet example.com

B. telnet example.com 443

C. telnet -p 443 example.com

D. telnet example.com:443

Solution: Explanation: The correct syntax is `telnet <host> <port>`. So `telnet example.com 443` tests if port 443 (HTTPS) is open. Modern alternatives include `nc -vz example.com 443` or PowerShell's `Test-NetConnection -Port 443`.

40. What is the purpose of `ssh-keygen`?

A. To connect to a remote server

B. To generate SSH key pairs for authentication

C. To copy files securely

D. To restart SSH service

Solution: Explanation: `ssh-keygen` generates SSH key pairs (public and private keys) for authentication. Key-based authentication is more secure than passwords and enables passwordless login. The command `ssh-keygen -t ed25519` generates a modern, secure key pair.

41. In the ping output, what does "mdev" or "stddev" represent?

A. Maximum deviation

B. Standard deviation of RTT (measures jitter/variability)

C. Minimum delay

D. Packet size

Solution: Explanation: The standard deviation (stddev or mdev) in ping output measures the variability or jitter in Round-Trip Time (RTT). Low values indicate stable latency, while high values suggest inconsistent network performance, possibly due to congestion or Wi-Fi interference.

42. Which combination of commands would correctly navigate to a directory, list its contents, and create a new subdirectory?

A. cd mydir && mkdir newdir && ls

B. cd mydir && ls && mkdir newdir

C. ls mydir && cd mydir && mkdir newdir

D. mkdir newdir && cd mydir && ls

Solution: Explanation: First `cd mydir` changes to the directory, then `ls` lists its contents, and finally `mkdir newdir` creates a new subdirectory. The `&&` operator ensures each command succeeds before executing the next.

43. When should you use an annotated Git tag instead of a lightweight tag?

A. For temporary testing

B. For local branches only

C. For official releases requiring metadata (tagger, date, message)

D. When you want to delete it later

Solution: Explanation: Annotated tags (created with `git tag -a v1.0 -m "Release v1.0"`) are full objects with metadata including tagger name, date, and message. They're recommended for release markers. Lightweight tags are just pointers and are suitable for temporary or personal markers.

44. In argparse, what is the difference between positional and optional arguments?

A. There is no difference

B. Positional arguments start with -

C. Positional arguments are required by position; optional arguments start with - or –

D. Optional arguments cannot have default values

Solution: Explanation: Positional arguments are required and identified by their position (e.g., `filename`). Optional arguments start with `-` or `--` (e.g., `--verbose`) and are not required unless specified with `required=True`. Optional arguments can appear in any order.

45. What is the correct workflow to push a new local branch to a remote repository?

A. git push origin

B. git commit -m ”message” && git push

C. git checkout -b new-branch && git push -u origin new-branch

D. git branch new-branch && git push

Solution: Explanation: The correct workflow is: 1) Create and switch to the new branch with `git checkout -b new-branch`, 2) Make commits, 3) Push with `git push -u origin new-branch`. The `-u` flag sets the upstream branch, allowing future pushes with just `git push`.

46. Which Python module provides the most secure way to handle command-line arguments with type validation and automatic help generation?

    A. sys

    B. os

    C. argparse

    D. input()

Solution: Explanation: `argparse` provides robust command-line argument parsing with automatic type conversion, validation, default values, and help generation. While `sys.argv` gives raw arguments as strings, `argparse` handles the complexity of parsing, validating, and generating user-friendly error messages and documentation.

End of Practice Set

Review your answers carefully and understand the explanations!